

FMCG Network Infrastructure Design

Part 01: CISCO ON-PREMISES SOLUTION

[IP Addressing, Subnetting, Calculations, Network Topology, Cabling, Device Selection, Cost Analysis]

Part 02: AWS VPC CLOUD NETWORK DESIGN

[AWS VPC Design, Subnet Mapping, And AWS Pricing Research and Cost Analysis]

Part 03: COMPARISON & RECOMMENDATION

[Comparing both solutions to find the best fit]

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Part 01: Cisco On-Premises Solution

1. Background

This section presents the Cisco on-premises network design for the FMCG client. The client requires a secure, scalable and durable network for nine departments: Marketing, Sales, HR, Technical, Customer Service, Financial, Strategy, Legal and Quality Control. Because the client also asked for possible expansion, the design must support both current users and future growth while maintaining a 10 Gbps-capable backbone.

The Cisco solution uses a private Class B address space, Variable Length Subnet Masking (VLSM), departmental VLAN segmentation and a hierarchical star topology. These choices improve IP address efficiency, simplify network management and support higher security by separating department traffic.

2. Department Host and Printer Requirements

The first step is to calculate how many IP addresses are needed for each department. Growth buffers were added exactly as needed in the project requirement: 10% for Sales and Quality Control, and 20% for Customer Service.

Printer allocation follows the rule of 1 printer for every 30 employees, with at least one printer for smaller departments which have fewer than 30 employees.

Department	Current Staff	Growth	Total Hosts Needed (staffs)	Printer Calculation**	Printers
Marketing	40	None	40	$40/30 = 1.33 > 2$	2
Sales	200	10%	$200 + 20 = 220$	$220/30 = 7.33 > 8$	8
HR	50	None	50	$50/30 = 1.67 > 2$	2
Technical	30	None	30	$30/30 = 1.00 > 1$	1
Customer Service	800	20%	$800 + 160 = 960$	$960/30 = 32.00 > 32$	32
Financial	20	None	20	Minimum 1 printer	1
Strategy	10	None	10	Minimum 1 printer	1
Legal	8	None	8	Minimum 1 printer	1
Quality Control	100	10%	$100 + 10 = 110$	$110/30 = 3.67 \rightarrow 4$	4
Total	1258	-	1448	-	52

* **For printer, we Always rounded up the fraction got from dividing number of employees by 30.

*Totals: 1,448 host devices and 52 printers (Total 1500 hosts) are required across all departments.

3. Class Selection and CIDR Justification

A private Class B range, 172.16.0.0/16 (16 CIDR is default mask for class B) is recommended.

The most important reason is the size of the Customer Service department. It requires 960 usable host addresses by itself.

Public IPv4 Recommendation:

While private Class B addressing (172.16.0.0/16) is used for internal network design, the organization will require public IP addresses from the ISP for external communication. A small Class C public IP block (e.g., /29 or /30) is recommended, as only edge devices such as routers or firewalls require public addressing. Internal devices will use private IPs with NAT (Network Address Translation) for Internet access.

Class Choice Explanation:

01. A Class A network leaves 24 host bits in the original /8 network. That provides $2^{24} = 16,777,216$ total addresses or 16,777,214 usable addresses (excluding network and broadcast addresses). This is far more than enough for most designs and allows extensive room for future subnetting and large-scale network growth. Class A provides excessive address space and is inefficient for this organization's size.

02. A Class B network leaves 16 host bits in the original /16 network. That provides $2^{16} = 65,536$ total addresses or 65,534 usable addresses (excluding network and broadcast addresses). This is more than enough for the current design and still leaves space for future subnet creation.

03. A Class C network leaves only 8 host bits, which gives $2^8 = 256$ total addresses, or 254 usable addresses (excluding network and broadcast addresses). That is not enough for 960 hosts.

4. Subnet Sizing Methodology (Including the +2 Rule (network ID and broadcast ID))

For accurate subnet sizing, two additional IP addresses must be reserved in every subnet: one for the Network ID and one for the Broadcast ID.

Therefore, the sizing check is not based only on the number of hosts; it is based on required IP addresses.

Sizing formula: Required IP addresses = Actual hosts + 2 (network ID and broadcast ID).

After that, the next suitable power of 2 is chosen, and the matching CIDR prefix is selected.

Department	Hosts Needed	Hosts + 2	Next Power of 2	Selected Prefix	Usable Hosts
Customer Service	960	962	1024	/22	1022
Sales	220	222	256	/24	254
Quality Control	110	112	128	/25	126
HR	50	52	64	/26	62
Marketing	40	42	64	/26	62
Technical	30	32	32	/27	30
Financial	20	22	32	/27	30
Strategy	10	12	16	/28	14
Legal	8	10	16	/28	14

Worked example for Customer Service:

960 hosts require 962 total addresses after adding Network ID and Broadcast ID. Since 512 is too small and 1024 is the next suitable block, 10 host bits are required. Therefore, the selected subnet is /22, which provides 1,022 usable host addresses.

CIDR note:

CIDR stands for Classless Inter-Domain Routing. Instead of thinking only of old classful sizes, we borrow host bits to create smaller subnets that match the department requirements. For example, a /22 subnet means out of 32 bits (IPv4) 22 bits are used for the network portion, and 10 bits are left for hosts.

5. Full VLSM and Subnet Calculation

Variable Length Subnet Masking (VLSM) is a network design technique that enhances IP address efficiency by allowing different-sized subnets within the same network, unlike **Fixed-Length Subnet Masking (FLSM)**. It enables assigning smaller, tailored subnet blocks to smaller departments, minimizing waste by calculating exact host requirements (e.g., /25 for 100 hosts, /29 for 6 hosts).

VLSM is used so that each department receives a subnet sized as closely as possible to its requirement. The departments are arranged from the largest host requirement to the smallest, then each department is assigned the next available network block.

General subnet formula: Usable hosts = $2^{(\text{host bits})} - 2$. The -2 (network ID and broadcast ID)

Step-by-step host-bit selection:

Customer Service needs 960 hosts. $2^9 - 2 = 510$, which is too small. $2^{10} - 2 = 1,022$, which is enough. Therefore, host bits = 10 and prefix = /22.

Sales need 220 hosts. $2^7 - 2 = 126$, too small. $2^8 - 2 = 254$, enough. Therefore, host bits = 8 and prefix = /24.

Quality Control needs 110 hosts. $2^6 - 2 = 62$, too small. $2^7 - 2 = 126$, enough. Therefore, host bits = 7 and prefix = /25.

HR needs 50 hosts. $2^5 - 2 = 30$, too small. $2^6 - 2 = 62$, enough. Therefore, host bits = 6 and prefix = /26.

Marketing needs 40 hosts. $2^5 - 2 = 30$, too small. $2^6 - 2 = 62$, enough. Therefore, host bits = 6 and prefix = /26.

Technical needs 30 hosts. $2^5 - 2 = 30$, enough exactly. Therefore, host bits = 5 and prefix = /27.

Financial needs 20 hosts. $2^4 - 2 = 14$, too small. $2^5 - 2 = 30$, enough. Therefore, host bits = 5 and prefix = /27.

Strategy needs 10 hosts. $2^3 - 2 = 6$, too small. $2^4 - 2 = 14$, enough. Therefore, host bits = 4 and prefix = /28.

Legal needs 8 hosts. $2^3 - 2 = 6$, too small. $2^4 - 2 = 14$, enough. Therefore, host bits = 4 and prefix = /28.

5.1 CIDR-to-Mask Conversion Table

Prefix	Subnet Mask	Host Bits details	Host Bits	Usable Hosts
/22	255.255.252.0	(8bits +2)	10	1,022
/24	255.255.255.0	Full 4 th Octet	8	254
/25	255.255.255.128	7bits from last Octet	7	126
/26	255.255.255.192	6bits from last Octet	6	62
/27	255.255.255.224	5bits from last Octet	5	30
/28	255.255.255.240	4bits from last Octet	4	14

(Calculation Note: Subnet mask values are calculated by subtracting the block size from 256 in the relevant octet (e.g., $256 - 64 = 192$ for /26). Or $256 (2^{8(\text{total Octet bits})}) - 2^{(\text{host bits})}$)

CIDR (Classless Inter-Domain Routing) allows flexible allocation of IP addresses by borrowing host bits to create subnets of different sizes. This enables efficient use of address space by matching subnet sizes to department requirements instead of using fixed class-based addressing.

5.2 Binary Example for CIDR Understanding

Example: /22 means that 22 bits belong to the network portion and 10 bits remain for hosts.

Subnet mask in dotted decimal: 255.255.252.0

Subnet mask in binary: 11111111.11111111.11111100.00000000

Host capacity check: $2^{10} = 1,024$ total addresses and $1,024 - 2 = 1,022$ usable addresses.

Note: This binary explanation shows exactly how the prefix length becomes the subnet mask and why the chosen subnet is large enough.

5.3 Detailed Subnet Allocation – Table A (Size Selection) with VLSM

Order	Department	Hosts Needed	Host Bits Needed	CIDR	Subnet Mask
1st	Customer Service	960	10	/22	255.255.252.0
2nd	Sales	220	8	/24	255.255.255.0
3rd	Quality Control	110	7	/25	255.255.255.128
4th	HR	50	6	/26	255.255.255.192
5th	Marketing	40	6	/26	255.255.255.192
6th	Technical	30	5	/27	255.255.255.224
7th	Financial	20	5	/27	255.255.255.224
8th	Strategy	10	4	/28	255.255.255.240
Last	Legal	8	4	/28	255.255.255.240

5.4 Detailed Subnet Allocation – Table B (Actual Network Ranges)

Department	Network ID	First Usable - Last Usable	Broadcast ID	Reason for next block
Customer Service	172.16.0.0	172.16.0.1 - 172.16.3.254	172.16.3.255	/22 uses block size 4 in the 3rd octet, so next network begins at 172.16.4.0
Sales	172.16.4.0	172.16.4.1 - 172.16.4.254	172.16.4.255	/24 uses entire 4th octet, so next network begins at 172.16.5.0
Quality Control	172.16.5.0	172.16.5.1 - 172.16.5.126	172.16.5.127	/25 uses block size 128, so next free network begins at 172.16.5.128
HR	172.16.5.128	172.16.5.129 - 172.16.5.190	172.16.5.191	/26 uses block size 64, so next free network begins at 172.16.5.192
Marketing	172.16.5.192	172.16.5.193 - 172.16.5.254	172.16.5.255	Remaining space in 172.16.5.x is completed; next network begins at 172.16.6.0
Technical	172.16.6.0	172.16.6.1 - 172.16.6.30	172.16.6.31	/27 uses block size 32, so next free network begins at 172.16.6.32
Financial	172.16.6.32	172.16.6.33 - 172.16.6.62	172.16.6.63	/27 block size again gives next network at 172.16.6.64
Strategy	172.16.6.64	172.16.6.65 - 172.16.6.78	172.16.6.79	/28 uses block size 16, so next network begins at 172.16.6.80
Legal	172.16.6.80	172.16.6.81 - 172.16.6.94	172.16.6.95	Final subnet in the shown VLSM plan

Why the first subnet is /22:

Customer Service is the largest department. A /22 subnet leaves 10 host bits. Therefore, $2^{10} = 1,024$ total addresses, and $1,024 - 2 = 1,022$ usable addresses. This safely covers 960 hosts and leaves a small spare capacity within the subnet.

5.5 Summary of Key Subnets (Example)

First Subnet (Customer Service):

Network ID: 172.16.0.0

Usable Range: 172.16.0.1 – 172.16.3.254

Broadcast ID: 172.16.3.255

Second Subnet (Sales):

Network ID: 172.16.4.0

Usable Range: 172.16.4.1 – 172.16.4.254

Broadcast ID: 172.16.4.255

Last Subnet (Legal):

Network ID: 172.16.6.80

Usable Range: 172.16.6.81 – 172.16.6.94

Broadcast ID: 172.16.6.95

Network increment explanation:

The /22 subnet mask is 255.255.252.0. The interesting octet is the third octet because 252 means a block size of 4 ($256 - 252 = 4$). So /22 networks in this range increase by 4 in the third octet: 172.16.0.0, 172.16.4.0, 172.16.8.0, and so on. Because Customer Service uses 172.16.0.0/22, the next free network starts at 172.16.4.0.

The /24 subnet mask is 255.255.255.0, so the block size is 1 in the fourth octet. That is why Sales is assigned 172.16.4.0/24, and the next available network after that is 172.16.5.0.

The /25 subnet mask is 255.255.255.128, so the block size is 128 (2nd bit from left) in the fourth octet. That creates 172.16.5.0/25 and 172.16.5.128/25. In this design, the first half is used for Quality Control, and the next free address begins at 172.16.5.128.

The /26 subnet mask is 255.255.255.192, so the block size is 64 (3rd bit from left) in the fourth octet. This creates subnets starting at .0, .64, .128 and .192. Therefore, HR is assigned 172.16.5.128/26 and Marketing is assigned 172.16.5.192/26.

The /27 subnet mask is 255.255.255.224, so the block size is 32 (4th bit from left) in the fourth octet. That is why Technical is 172.16.6.0/27 and Financial is 172.16.6.32/27.

The /28 subnet mask is 255.255.255.240, so the block size is 16 (5th bit from left) in the fourth octet. That is why Strategy is 172.16.6.64/28 and Legal is 172.16.6.80/28.

5.6 Address Utilization and Wastage Analysis

Department	Usable Hosts Available	Hosts Used	Unused / Spare Addresses
Customer Service	1022	960	62
Sales	254	220	34
Quality Control	126	110	16
HR	62	50	12
Marketing	62	40	22
Technical	30	30	0
Financial	30	20	10
Strategy	14	10	4
Legal	14	8	6

This utilization table shows why VLSM is preferred over equal size subnetting. Each department receives a subnet that is close to its real requirement, which reduces IP waste while still leaving a modest growth margin inside most subnets.

6. VLAN, Topology and Inter-VLAN Routing Notes

The physical topology recommended for this design is hierarchical extended star topology. Each department connects to an access switch, the access switches connect to distribution/core switches, and the core connects to the router and WAN edge. This topology supports durability because if one access switch fails, the rest of the network continues operating.

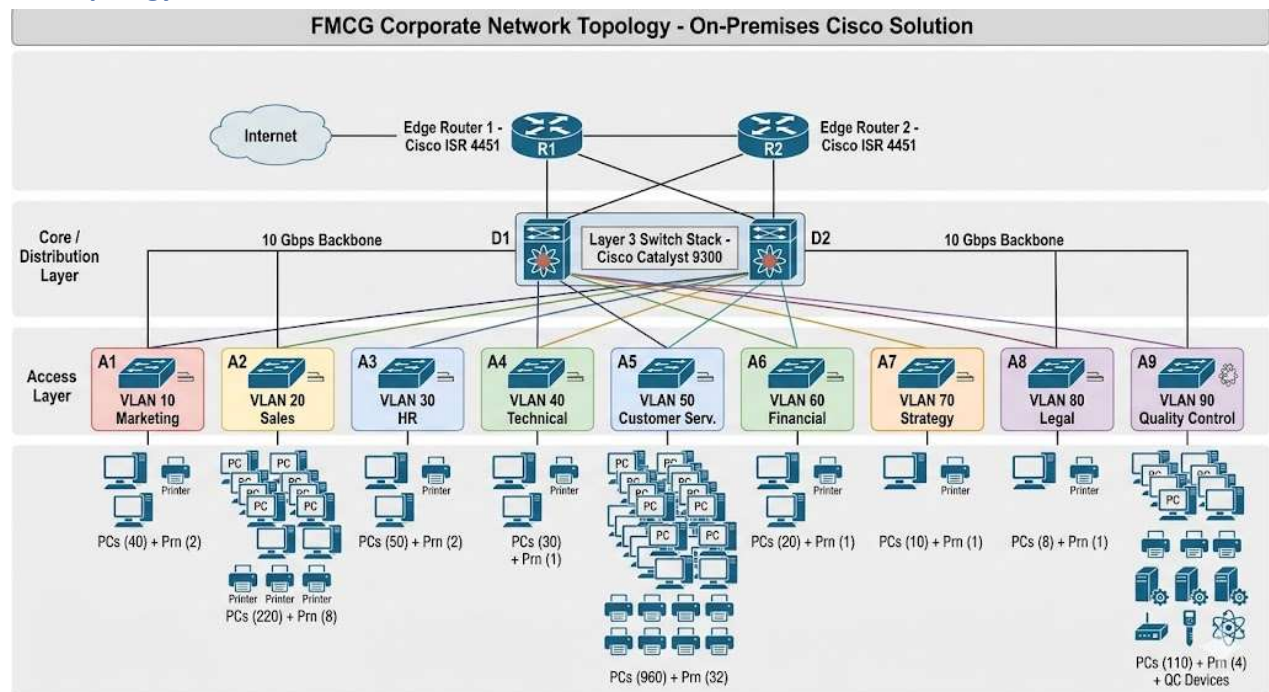
Department	Suggested VLAN	Reason
Marketing	VLAN 10	Separate marketing traffic and printers
Sales	VLAN 20	Large user group with future growth
HR	VLAN 30	Sensitive employee records
Technical	VLAN 40	IT/admin systems
Customer Service	VLAN 50	Largest department subnet
Financial	VLAN 60	High security for finance systems
Strategy	VLAN 70	Management/strategy systems
Legal	VLAN 80	Confidential legal documents
Quality Control	VLAN 90	Operational and QC devices

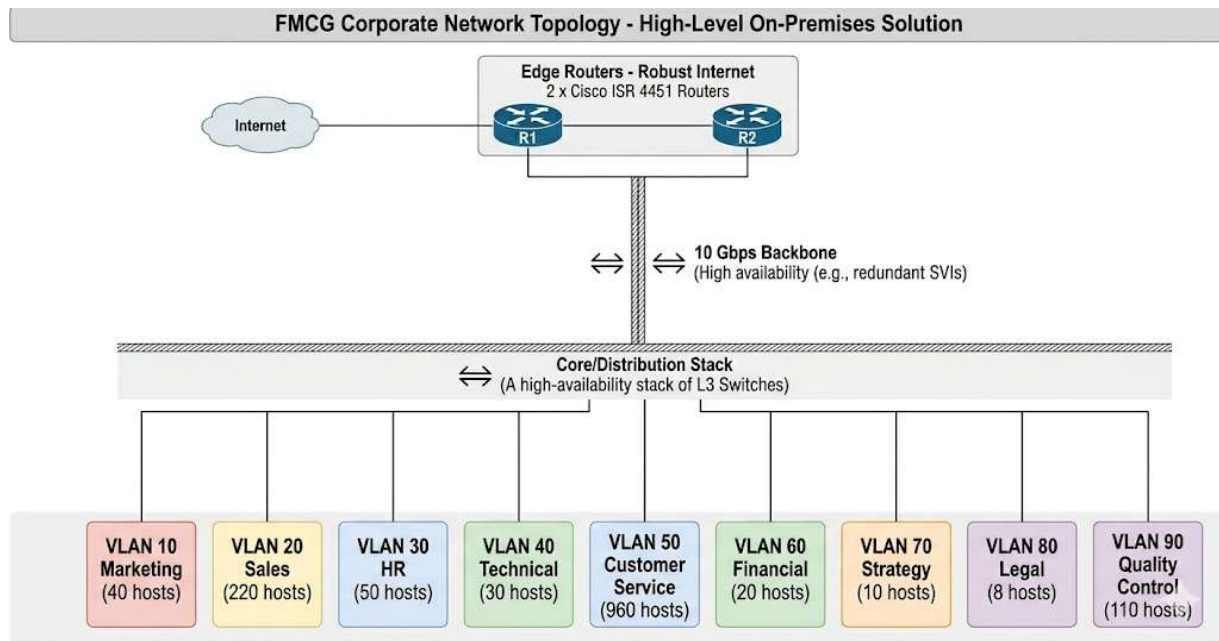
Inter-VLAN routing should be enabled on a Layer 3 switch or on the router, depending on the final implementation. In a switched campus design, the most common approach is to create one SVI (Switch Virtual Interface) per VLAN and use that SVI address as the default gateway for hosts in that department.

Example gateway idea: VLAN 10 (Marketing) can use 172.16.5.193 as the subnet gateway, VLAN 20 (Sales) can use 172.16.4.1, and VLAN 50 (Customer Service) can use 172.16.0.1. Access control lists can then be applied to limit which VLANs can communicate with highly sensitive departments such as Finance, HR and Legal.

The logical topology is VLAN-based segmentation. Each department should be placed in its own VLAN so traffic can be isolated and controlled using ACLs and inter-VLAN routing policies. This is especially important for Finance, HR and Legal because those departments may handle confidential information.

6.1 Topology





Designed with the help of Network tools.

Explanation:

This topology uses a hierarchical extended star design. All departmental access switches connect to a central core/distribution Layer 3 switch, which then connects to the edge routers and the Internet/WAN. Each department is separated into its own VLAN for better security, management, and performance. Inter-VLAN routing is handled at the Layer 3 switch using SVIs as default gateways. This design improves scalability, fault isolation, and supports future expansion.

For labels, use:

- Marketing — VLAN 10 — 172.16.5.192/26
- Sales — VLAN 20 — 172.16.4.0/24
- HR — VLAN 30 — 172.16.5.128/26
- Technical — VLAN 40 — 172.16.6.0/27
- Customer Service — VLAN 50 — 172.16.0.0/22
- Financial — VLAN 60 — 172.16.6.32/27
- Strategy — VLAN 70 — 172.16.6.64/28
- Legal — VLAN 80 — 172.16.6.80/28
- Quality Control — VLAN 90 — 172.16.5.0/25

This hierarchical extended star topology provides high scalability, fault isolation, and easier troubleshooting. If one access switch fails, it does not impact the entire network. It also supports future expansion by allowing additional switches to be added without major redesign.

Network Address Translation (NAT) will be implemented at the edge router to allow internal private IP addresses to communicate with external networks using public IP addresses.

6.2 Cabling Design

To support a 10 Gbps network, the following cabling is recommended:

Backbone connections: Fiber optic cables (OM3/OM4 multimode or single mode) for high-speed and long-distance communication.

Access layer connections: Category 6a (Cat6a) Ethernet cables, which support up to 10 Gbps over standard office distances.

This combination ensures high performance, reliability, and future scalability of the network.

7. Detailed Budget and 10-Year TCO Calculation

The earlier draft only listed total amounts. This section shows the full cost calculation process. Because exact pricing changes by supplier and date, the figures below are presented as reasonable estimate values in CAD for assignment purposes. The same formulas can be reused if you later replace the unit prices with vendor quotations.

7.1 Hardware Cost Table

Item	Model / Description	Qty	Unit Cost (CAD)	Calculation	Line Total (CAD)
Access / Distribution Switch	Cisco Catalyst 9300 Series	10	6,000	10 x 6,000	60,000
Router	Cisco ISR 4451	2	8,000	2 x 8,000	16,000

Assumptions used in the estimate:

Nine departments are served each by at least one access switch, and one additional switch is reserved for aggregation/distribution.

Total devices required: 10 switches and 2 routers to support all departments and ensure redundancy.

7.2 Labor Cost Table

Role	Hours	Rate (CAD/hr)	Cost (CAD)
Network design, installation and testing	80	60	4,800

Note:

Labor cost is CAD 60 per hour as stated in the project instructions.

Annual maintenance is estimated as 5% of total hardware cost.

7.3 Maintenance and 10-Year TCO Table

Hardware subtotal calculation:

Hardware Cost = $(10 \times \text{CAD } 6,000) + (2 \times \text{CAD } 8,000)$

Hardware Cost = CAD 60,000 + CAD 16,000

Hardware Cost = CAD 76,000

labor subtotal calculation:

Implementation labor = 80 hours x CAD 60/hour

Implementation labor = CAD 4,800

Component	Formula	Amount (CAD)
Hardware	$(10 \times 6,000) + (2 \times 8,000)$	76,000
Labour	80×60	4,800
Maintenance over 10 years	$(76,000 \times 0.05) \times 10$	38,000
Total Cost of Ownership	$76,000 + 4,800 + 38,000$	118,800

Maintenance calculation:

Annual Maintenance = $5\% \times \text{Hardware Cost}$

Annual Maintenance = $0.05 \times 76,000$

Annual Maintenance = CAD 3,800

10-Year Maintenance = Annual Maintenance x 10

10-Year Maintenance = $3,800 \times 10$

10-Year Maintenance = CAD 38,000

Budget justification note:

Cisco Catalyst 9300 switches were selected due to their support for VLAN segmentation, Layer 3 capabilities, and 10 Gbps uplinks suitable for enterprise environments. Cisco ISR 4451 routers were chosen because they provide reliable WAN connectivity, advanced routing features, and support for secure network operations such as VPN and NAT.

Pricing Note:

The device prices used in this estimate are based on average market pricing for Cisco enterprise equipment (Cisco official partners, CDW, and online enterprise hardware vendors). Actual pricing may vary depending on supplier, configuration, and discounts.

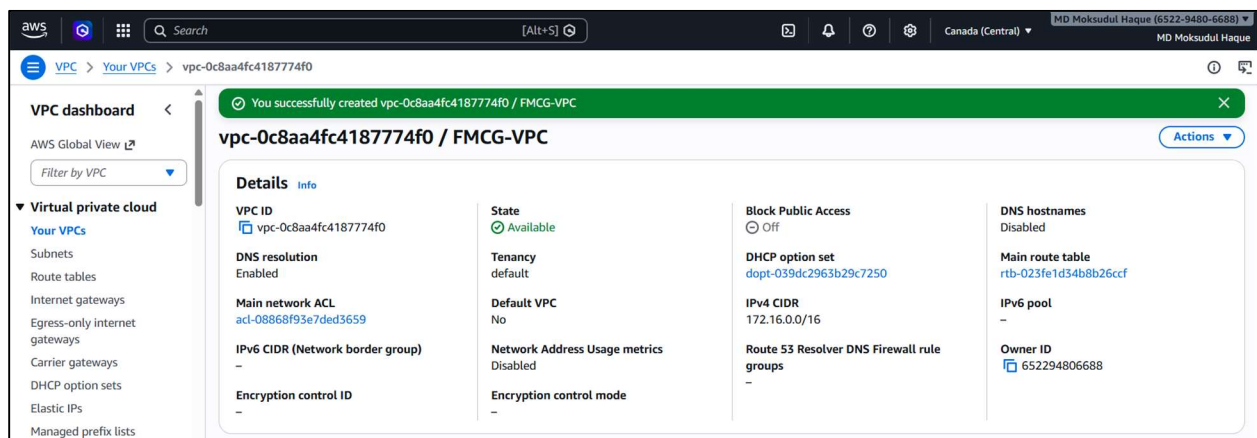
Part 02: AWS VPC CLOUD NETWORK DESIGN

1.0 Introduction

This section presents the AWS cloud-based network design for the FMCG organization. Instead of using physical infrastructure, the network is implemented using Amazon Virtual Private Cloud (VPC), which provides a scalable, secure and highly available virtual networking environment.

The design follows the same logical structure as the Cisco network, where each department is isolated using subnets. AWS services such as EC2, Internet Gateway, Route Tables and Security Groups are used to replicate enterprise-level networking in the cloud.

1.2 VPC Created



A /16 CIDR block is selected to match the Cisco design and allow sufficient address space for all departments and future expansion. This ensures consistency between on-premises and cloud architecture.

2.1 SUBNET DESIGN

Department	CIDR	Subnet Type
Customer Service	172.16.0.0/22	Private
Sales	172.16.4.0/24	Private
Quality Control	172.16.5.0/25	Private
HR	172.16.5.128/26	Private
Marketing	172.16.5.192/26	Public
Technical	172.16.6.0/27	Private
Financial	172.16.6.32/27	Private
Strategy	172.16.6.64/28	Private
Legal	172.16.6.80/28	Private

2.2 Subnet Creation (AWS)

You have successfully created 1 subnet: subnet-0f090e0aeebeda393

Subnets (9) Info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

< 1 >

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Customer Service	subnet-0b74dfb36da962650	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Sales	subnet-08c7f2cf84c35cce5	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Quality Control	subnet-04fb0cb83e8ad007f	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	HR	subnet-0af6797badbe490f8	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Marketing	subnet-0bb0e8d25492c2fa7	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Technical	subnet-0a527b8d7c9ed7078	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Financial	subnet-0ba142e0d88f5eeb4	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Strategy	subnet-048510e24b4fc14d0	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24
☐	Legal	subnet-0f090e0aeebeda393	Available	vpc-0c8aa4fc4187774f0 FMCG-VPC	Off	172.16.0.0/24

The public subnet is used by Marketing to host public-facing services. All other departments are placed in private subnets for enhanced security.

3.0 Internet Connectivity Design - Internet Gateway Created

aws

Search

[Alt+S]

Canada (Central)

MD Moksudul Haque (6522-9480-6688)

MD Moksudul Haque

VPC > Internet gateways > igw-042a3d5d32df9dc77

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

The following internet gateway was created: igw-042a3d5d32df9dc77 - FMCG-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

igw-042a3d5d32df9dc77 / FMCG-IGW

Actions

Details Info

Internet gateway ID

igw-042a3d5d32df9dc77

State

Detached

VPC ID

-

Owner

652294806688

Tags (1)

Search tags

Manage tags

< 1 >

Key	Value
Name	FMCG-IGW

3.1 Attach to VPC (FMCG-VPC)

Internet gateway igw-042a3d5d32df9dc77 successfully attached to vpc-0c8aa4fc4187774f0

Notifications 0 0 2 0 0

igw-042a3d5d32df9dc77 / FMCG-IGW

Actions

Details Info

Internet gateway ID

igw-042a3d5d32df9dc77

State

Attached

VPC ID

[vpc-0c8aa4fc4187774f0 | FMCG-VPC](#)

Owner

652294806688

Tags (1)

Search tags

Manage tags

< 1 >

Key	Value
Name	FMCG-IGW

4.0 Public Route Table Created

Route table rtb-03bea97011bc1be70 | Public-Route was created successfully.

rtb-03bea97011bc1be70 / Public-Route

Details

Route table ID

rtb-03bea97011bc1be70

VPC

vpc-0c8aa4fc4187774f0 | FMCG-VPC

Main

No

Explicit subnet associations

-

Edge associations

-

Owner ID

652294806688

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
172.16.0.0/16	local	Active	No	Create Route Table

Edit routes

Destination

172.16.0.0/16

Target

local

local

Internet Gateway

igw-042a3d5d32df9dc77

Status

Active

-

Propagated

No

No

Route Origin

CreateRouteTable

CreateRoute

Add route

Remove

Cancel

Preview

Save changes

4.1 Adding Internet Route

Updated routes for rtb-03bea97011bc1be70 / Public-Route successfully

Details

You have successfully created 1 route: 0.0.0.0/0.

rtb-03bea97011bc1be70 / Public-Route

Details

Route table ID

rtb-03bea97011bc1be70

VPC

vpc-0c8aa4fc4187774f0 | FMCG-VPC

Main

No

Explicit subnet associations

-

Edge associations

-

Owner ID

652294806688

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-042a3d5d32df9dc77	Active	No	Create Route
172.16.0.0/16	local	Active	No	Create Route Table

5.0 Associating Marketing Subnet

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/9)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	Customer Service	subnet-0b74dfb36da962650	172.16.0.0/22	-	Main (rtb-023fe1d34b8b26ccf)
☐	Sales	subnet-08c7f2cf84c35ce5	172.16.4.0/24	-	Main (rtb-023fe1d34b8b26ccf)
☐	Quality Control	subnet-04fb0cb83e8ad007f	172.16.5.0/25	-	Main (rtb-023fe1d34b8b26ccf)
☐	HR	subnet-0af6797badbe490f8	172.16.5.128/26	-	Main (rtb-023fe1d34b8b26ccf)
☒	Marketing	subnet-0bb0e8d25492c2fa7	172.16.5.192/26	-	Main (rtb-023fe1d34b8b26ccf)
☐	Technical	subnet-0a527b8d7c9ed7078	172.16.6.0/27	-	Main (rtb-023fe1d34b8b26ccf)
☐	Financial	subnet-0ba142e0d88f5eeb4	172.16.6.32/27	-	Main (rtb-023fe1d34b8b26ccf)
☐	Strategy	subnet-048510e24b4fc14d0	172.16.6.64/28	-	Main (rtb-023fe1d34b8b26ccf)
☐	Legal	subnet-0f090e0aeebeda393	172.16.6.80/28	-	Main (rtb-023fe1d34b8b26ccf)

Selected subnets

subnet-0bb0e8d25492c2fa7 / Marketing

You have successfully updated subnet associations for rtb-03bea97011bc1be70 / Public-Route.

rtb-03bea97011bc1be70 / Public-Route

Actions

Details

Route table ID

rtb-03bea97011bc1be70

VPC

vpc-0c8aa4fc4187774f0 | FMCG-VPC

Main

No

Owner ID

652294806688

Explicit subnet associations

subnet-0bb0e8d25492c2fa7 / Marketing

Edge associations

-

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-042a3d5d32df9dc77	Active	No	Create Route
172.16.0.0/16	local	Active	No	Create Route Table

5.1 Enable Public IP from Edit subnet settings

You have successfully changed subnet settings:

- Enable auto-assign public IPv4 address

Edit subnet settings

Subnet

Subnet ID

subnet-0bb0e8d25492c2fa7

Name

Marketing

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address

☐ Enable auto-assign customer-owned IPv4 address

6.0 Private Route Table Created

Route table rtb-02fea86e3ae2d0c9f | Private-Route was created successfully.

Actions

rtb-02fea86e3ae2d0c9f / Private-Route

Details

Route table ID
rtb-02fea86e3ae2d0c9f

VPC
vpc-0c8aa4fc4187774f0 | FMCG-VPC

Main
No

Owner ID
652294806688

Explicit subnet associations
-

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

Filter routes

1

Destination

Target

Status

Propagated

Route Origin

172.16.0.0/16

local

Active

No

Create Route Table

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (8/9)

Filter subnet associations

1

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Customer Service	subnet-0b74dfb36da962650	172.16.0.0/22	-	Main (rtb-023fe1d34b8b26ccf)
☒	Sales	subnet-08c7f2cf84c35cce5	172.16.4.0/24	-	Main (rtb-023fe1d34b8b26ccf)
☒	Quality Control	subnet-04fb0cb83e8ad007f	172.16.5.0/25	-	Main (rtb-023fe1d34b8b26ccf)
☒	HR	subnet-0af6797badbe490f8	172.16.5.128/26	-	Main (rtb-023fe1d34b8b26ccf)
☐	Marketing	subnet-0bb0e8d25492c2fa7	172.16.5.192/26	-	rtb-03bea97011bc1be70 / Public-Route
☒	Technical	subnet-0a527b8d7c9ed7078	172.16.6.0/27	-	Main (rtb-023fe1d34b8b26ccf)
☒	Financial	subnet-0ba142e0d88f5eeb4	172.16.6.32/27	-	Main (rtb-023fe1d34b8b26ccf)
☒	Strategy	subnet-048510e24b4fc14d0	172.16.6.64/28	-	Main (rtb-023fe1d34b8b26ccf)
☒	Legal	subnet-0f090e0aeebeda393	172.16.6.80/28	-	Main (rtb-023fe1d34b8b26ccf)

Selected subnets

subnet-0b74dfb36da962650 / Customer Service

subnet-08c7f2cf84c35cce5 / Sales

subnet-0af6797badbe490f8 / HR

subnet-04fb0cb83e8ad007f / Quality Control

subnet-0a527b8d7c9ed7078 / Technical

subnet-0ba142e0d88f5eeb4 / Financial

subnet-048510e24b4fc14d0 / Strategy

subnet-0f090e0aeebeda393 / Legal

6.1 Private Route Table Subnet Association

You have successfully updated subnet associations for rtb-02fea86e3ae2d0c9f / Private-Route.

Actions

rtb-02fea86e3ae2d0c9f / Private-Route

Details

Route table ID
rtb-02fea86e3ae2d0c9f

VPC
vpc-0c8aa4fc4187774f0 | FMCG-VPC

Main
No

Owner ID
652294806688

Explicit subnet associations
8 subnets

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (8)

Edit subnet associations

Find subnet association

1

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Customer Service	subnet-0b74dfb36da962650	172.16.0.0/22	-
Sales	subnet-08c7f2cf84c35cce5	172.16.4.0/24	-
Quality Control	subnet-04fb0cb83e8ad007f	172.16.5.0/25	-
HR	subnet-0af6797badbe490f8	172.16.5.128/26	-
Technical	subnet-0a527b8d7c9ed7078	172.16.6.0/27	-
Financial	subnet-0ba142e0d88f5eeb4	172.16.6.32/27	-
Strategy	subnet-048510e24b4fc14d0	172.16.6.64/28	-
Legal	subnet-0f090e0aeebeda393	172.16.6.80/28	-

7.0 NAT Gateway Created

NAT gateway nat-1eb98cf6473bcf60a / FMCG-NAT was created successfully.

nat-1eb98cf6473bcf60a / FMCG-NAT

Actions

Details

NAT gateway ID
 nat-1eb98cf6473bcf60a

Availability mode
Regional

State
 Pending

State message [Info](#)
-

NAT gateway ARN
 arn:aws:ec2:ca-central-1:652294806688:natgateway/nat-1eb98cf6473bcf60a

Connectivity type
Public

Created
 Friday, April 17, 2026 at 12:39:28 CST

Deleted
-

VPC
[vpc-0c8aa4fc4187774f0 / FMCG-VPC](#)

Method of EIP allocation
Automatic

[Alt+S]

Canada (Central)

MD Moksubul Haque (6522-9480-6688)

VPC > NAT gateways > Create NAT gateway

Elastic IP address 16.54.19.215 (eipalloc-04c966d93b6935d48) allocated.

NAT gateway nat-03cb62e9558934383 / FMCG-NAT was created successfully.

nat-03cb62e9558934383 / FMCG-NAT

Actions

Details

NAT gateway ID
 nat-03cb62e9558934383

Connectivity type
Public

State
 Pending

State message [Info](#)
-

NAT gateway ARN
 arn:aws:ec2:ca-central-1:652294806688:natgateway/nat-03cb62e9558934383

Primary public IPv4 address
-

Primary private IPv4 address
-

Primary network interface ID
-

VPC
[vpc-0c8aa4fc4187774f0 / FMCG-VPC](#)

Subnet
[subnet-0bb0e8d25492c2fa7 / Marketing](#)

Created
 Friday, April 17, 2026 at 12:56:27 CST

Deleted
-

A zonal NAT Gateway is deployed inside the Marketing public subnet to allow private subnets to securely access the internet.

7.1 Route table update:

Edit routes

Destination	Target	Status	Propagated	Route Origin
172.16.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute
	nat-03cb62e9558934383			

Add route

Private subnets route internet traffic through the NAT Gateway rather than directly to the Internet Gateway, ensuring security.

Updated routes for rtb-02fea86e3ae2d0c9f / Private-Route successfully
Details
You have successfully created 1 route: 0.0.0.0/0.

rtb-02fea86e3ae2d0c9f / Private-Route

Actions

Details

Route table ID

rtb-02fea86e3ae2d0c9f

VPC

vpc-0c8aa4fc4187774f0 | FMCG-VPC

Main

No

Owner ID

652294806688

Explicit subnet associations

8 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-03cb62e9558934383	Active	No	Create Route
172.16.0.0/16	local	Active	No	Create Route Table

8.0 EC2 Deployment

EC2 instances are deployed within the VPC to simulate departmental systems. Each department is assigned an EC2 instance inside its respective subnet. The Marketing department is placed in a public subnet with a public IP, while all other departments are deployed in private subnets without direct internet access. These private instances access the internet through a NAT Gateway for updates and external communication.

Department	Subnet Type	EC2 Access
Marketing	Public Subnet	Public EC2 (with public IP)
Customer Service	Private Subnet	Private EC2
Sales	Private Subnet	Private EC2
Quality Control	Private Subnet	Private EC2
HR	Private Subnet	Private EC2
Technical	Private Subnet	Private EC2
Financial	Private Subnet	Private EC2
Strategy	Private Subnet	Private EC2
Legal	Private Subnet	Private EC2

Public EC2 instances are directly reachable from the internet, while private EC2 instances are protected and only accessible within the VPC or through secure methods such as a bastion host or VPN.

9.0 Key Pair Created

Successfully created key pair

Key pairs (1) [Info](#)

Actions

Create key pair

☐	Name	Type	Created	Fingerprint	ID
☐	FMCG-Key	rsa	2026/04/18 16:54 GMT-6	7b:cc:26:9f:21:59:e7:31:7c:cc:26:27:77:f9:c3:3...	key-0edae...

9.1 Public Security Group Created

Security group (sg-042a31c1485b50e6a | Public-SG) was created successfully

Details

- Creating security group
- Create 3 inbound rules
 - TCP : 22 : 71.17.25.72/32
 - TCP : 80 : 0.0.0.0/0
 - TCP : 443 : 0.0.0.0/0

sg-042a31c1485b50e6a - Public-SG

Actions

Details

Security group name

Public-SG

Security group ID

sg-042a31c1485b50e6a

Description

Public access for Marketing EC2

VPC ID

vpc-0c8aa4fc4187774f0

Owner

652294806688

Inbound rules count

3 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

Manage tags

Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0683bc901fc5a5bf7	IPv4	SSH	TCP	22	71.17.25.72/32	-
☐	-	sgr-098f7cee44e2c700a	IPv4	HTTP	TCP	80	0.0.0.0/0	-
☐	-	sgr-0b90274de9c72bbd4	IPv4	HTTPS	TCP	443	0.0.0.0/0	-

9.2 Private Security Group Created

Security group (sg-0f37423f551e6d6f7 | Private-SG) was created successfully

Details

- Creating security group
- Create 2 inbound rules
 - TCP : 22 : sg-042a31c1485b50e6a
 - All traffic : All : 172.16.0.0/16

sg-0f37423f551e6d6f7 - Private-SG

Actions

Details

Security group name

Private-SG

Security group ID

sg-0f37423f551e6d6f7

Description

Private access only

VPC ID

vpc-0c8aa4fc4187774f0

Owner

652294806688

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-0bf5c50763ce8191e	IPv4	All traffic	All	All	172.16.0.0/16
☐	-	sgr-0bd95a93ab18a0ca0	-	SSH	TCP	22	sg-042a31c1485b50e6...

9.3 Launch Public EC2 (Marketing)

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-0268ba3ee751775dc)

Launch log

Initializing requests

Succeeded

Launch initiation

Succeeded

Instance summary for i-0268ba3ee751775dc (Marketing-Server)

Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-0268ba3ee751775dc

IPv6 address

-

Hostname type

IP name: ip-172-16-5-247.ca-central-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

52.60.237.241 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

52.60.237.241 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-16-5-247.ca-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0c8aa4fc4187774f0 (FMCG-VPC)

Subnet ID

subnet-0bb0e8d25492c2fa7 (Marketing)

Instance ARN

arn:aws:ec2:ca-central-1:652294806688:instance/i-0268ba3ee751775dc

Private IPv4 addresses

172.16.5.247

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

-

Security groups

sg-042a31c1485b50e6a (Public-SG)

Owner ID

652294806688

Launch time

Sat Apr 18 2026 17:11:15 GMT-0600 (Central Standard Time)

▼ Inbound rules

Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source	Security groups	D
-	sgr-0683bc901fc5a5bf7	22	TCP	71.17.25.72/32	Public-SG	-
-	sgr-098f7cee44e2c700a	80	TCP	0.0.0.0/0	Public-SG	-
-	sgr-0b90274de9c72bbd4	443	TCP	0.0.0.0/0	Public-SG	-

▼ Outbound rules

Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	D
-	sgr-0c2fe065b0f48d9d6	All	All	0.0.0.0/0	Public-SG	-

26 | Page

✓ Inbound security group rules successfully modified on security group (sg-042a31c1485b50e6a | Public-SG)

▼ Details

- Update
 - TCP : 22 : 0.0.0.0/0

sg-042a31c1485b50e6a - Public-SG

Actions ▼

Details

Security group name
Public-SG

Security group ID
sg-042a31c1485b50e6a

Description
Public access for Marketing EC2

VPC ID
vpc-0c8aadfc4187774f0

Owner
652294806688

Inbound rules count
3 Permission entries

Outbound rules count
1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (3)

Manage tags Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-0683bc901fc5a5bf7	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-098f7cee44e2c700a	IPv4	HTTP	TCP	80	0.0.0.0/0
☐	-	sgr-0b90274de9c72bbd4	IPv4	HTTPS	TCP	443	0.0.0.0/0

9.4 Launch Private EC2 (Customer Service)

✓ Success
Successfully initiated launch of instance (i-024d99cbaaf9f5943)

Instance summary for i-024d99cbaaf9f5943 (CS-Server) Info

Connect Instance state ▼ Actions ▼

Updated less than a minute ago

Instance ID
i-024d99cbaaf9f5943

IPv6 address
-

Hostname type
IP name: ip-172-16-0-126.ca-central-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
-

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
-

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-16-0-126.ca-central-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0c8aadfc4187774f0 (FMCG-VPC)

Subnet ID
subnet-0b74dfb36da962650 (Customer Service)

Instance ARN
arn:aws:ec2:ca-central-1:652294806688:instance/i-024d99cbaaf9f5943

Private IPv4 addresses
172.16.0.126

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. Learn more

Auto Scaling Group name
-

Managed
false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Security details

IAM role
-

Security groups
sg-0f37423f51e6d5f7 (Private-SG)

Owner ID
652294806688

Launch time
Sat Apr 18 2026 17:15:05 GMT-0600 (Central Standard Time)

▼ Inbound rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0bf5c50763ce8191e	All	All	172.16.0.0/16	Private-SG	-
-	sgr-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	-

▼ Outbound rules

<

1

>

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sgr-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	-

9.5 Launch Private EC2 (Sales)

Success

Successfully initiated launch of Instance (i-0a54368e666a0886d)

Instance summary for i-0a54368e666a0886d (Sales-Server) Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-0a54368e666a0886d

IPv6 address

-

Hostname type

IP name: ip-172-16-4-51.ca-central-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

-

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-16-4-51.ca-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0c8aadfc4187774f0 (FMCG-VPC)

Subnet ID

subnet-08c7f2cf84c35cce5 (Sales)

Instance ARN

arn:aws:ec2:ca-central-1:652294806688:instance/i-0a54368e666a0886d

Private IPv4 addresses

172.16.4.51

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

-

Security groups

sg-0f37423f551e6d6f7 (Private-SG)

Owner ID

652294806688

Launch time

Sat Apr 18 2026 17:17:36 GMT-0600 (Central Standard Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0bf5c50765ce8191e	All	All	172.16.0.0/16	Private-SG	-
-	sgr-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	-

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sgr-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	-

9.6 Launch Private EC2 (HR)

Success

Successfully initiated launch of instance (i-07a56719e351470a2)

Instance summary for i-07a56719e351470a2 (HR-Server) Info

Updated less than a minute ago

Connect

Instance state

Actions

Instance ID

i-07a56719e351470a2

IPv6 address

—

Hostname type

IP name: ip-172-16-5-183.ca-central-1.compute.internal

Answer private resource DNS name

—

Auto-assigned IP address

—

IAM role

—

IMDSv2

Required

Operator

—

Public IPv4 address

—

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-16-5-183.ca-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0c8aa4fc4187774f0 (FMCG-VPC)

Subnet ID

subnet-0af6797badbe490f8 (HR)

Instance ARN

arn:aws:ec2:ca-central-1:652294806688:instance/i-07a56719e351470a2

Private IPv4 addresses

172.16.5.183

Public DNS

—

Elastic IP addresses

—

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

—

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

—

Security groups

sg-0f37423f551e6d6f7 (Private-SG)

Owner ID

652294806688

Launch time

Sat Apr 18 2026 17:18:32 GMT-0600 (Central Standard Time)

▼ Inbound rules

Q Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
—	sgr-0bf5c50763ce8191e	All	All	172.16.0.0/16	Private-SG	—
—	sgr-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	—

▼ Outbound rules

Q Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
—	sgr-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	—

✔ Success
Successfully initiated launch of instance [\(i-088cab98a65d6a78c\)](#)

9.8 Launch Private EC2 (Finance)

Success

Successfully initiated launch of instance (i-04258a87840f53cf8)

Instance summary for i-04258a87840f53cf8 (Finance-Server) Info

Updated less than a minute ago

Instance ID

i-04258a87840f53cf8

IPv6 address

—

Hostname type

IP name: ip-172-16-6-46.ca-central-1.compute.internal

Answer private resource DNS name

—

Auto-assigned IP address

—

IAM role

—

IMDSv2

Required

Operator

—

Public IPv4 address

—

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-16-6-46.ca-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0c8aa4fc4187774f0 (FMCG-VPC)

Subnet ID

subnet-0ba142e0d88f5eeb4 (Financial)

Instance ARN

arnaws:ec2:ca-central-1:652294806688:instance/i-04258a87840f53cf8

Private IPv4 addresses

172.16.6.46

Public DNS

—

Elastic IP addresses

—

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

—

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

—

Security groups

sg-0f37423f551e6d5f7 (Private-SG)

Owner ID

652294806688

Launch time

Sat Apr 18 2026 17:20:37 GMT-0600 (Central Standard Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
—	sgr-0bf5c50763ce8191e	All	All	172.16.0.0/16	Private-SG	—
—	sgr-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	—

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
—	sgr-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	—

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9.9 Launch Private EC2 (Strategy)

Success

Successfully initiated launch of instance (i-04a9ce42c5e819178)

Instance summary for i-04a9ce42c5e819178 (Strategy-Server) [Info](#)

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-04a9ce42c5e819178

IPv6 address

--

Hostname type

IP name: ip-172-16-6-77.ca-central-1.compute.internal

Answer private resource DNS name

--

Auto-assigned IP address

--

IAM role

--

IMDSv2

Required

Operator

--

Public IPv4 address

--

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-16-6-77.ca-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0c8aa4fc4187774f0 (FMCG-VPC)

Subnet ID

subnet-048510e24b4fc14d0 (Strategy)

Instance ARN

arn:aws:ec2:ca-central-1:652294806688:instance/i-04a9ce42c5e819178

Private IPv4 addresses

172.16.6.77

Public DNS

--

Elastic IP addresses

--

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Auto Scaling Group name

--

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

--

Security groups

sg-0f37423f551e6d6f7 (Private-SG)

Owner ID

652294806688

Launch time

Sat Apr 18 2026 17:21:30 GMT-0600 (Central Standard Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
--	sgr-0bf5c50763ce8191e	All	All	172.16.0.0/16	Private-SG	--
--	sgr-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	--

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
--	sgr-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	--

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9.10 Launch Private EC2 (Legal)

Success

Successfully initiated launch of instance (i-06cbeec50c967cb0d)

Instance summary for i-06cbeec50c967cb0d (Legal-Server) Info

Updated less than a minute ago

Instance ID

i-06cbeec50c967cb0d

IPv6 address

--

Hostname type

IP name: ip-172-16-6-88.ca-central-1.compute.internal

Answer private resource DNS name

--

Auto-assigned IP address

--

IAM role

--

IMDSv2

Required

Operator

--

Public IPv4 address

--

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-16-6-88.ca-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0c8aa4fc4187774f0 (FMCG-VPC)

Subnet ID

subnet-0f090e0aeebeda393 (Legal)

Instance ARN

arn:aws:ec2:ca-central-1:652294806688:instance/i-06cbeec50c967cb0d

Private IPv4 addresses

172.16.6.88

Public DNS

--

Elastic IP addresses

--

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

--

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

--

Security groups

sg-0f37423f551e6dsf7 (Private-SG)

Owner ID

652294806688

Launch time

Sat Apr 18 2026 17:22:45 GMT-0600 (Central Standard Time)

▼ Inbound rules

Q Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
--	sg-0bf5c50763ce8191e	All	All	172.16.0.0/16	Private-SG	--
--	sg-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	--

▼ Outbound rules

Q Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
--	sg-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	--

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9.11 Launch Private EC2 (Quality Control)

 **Success**
Successfully initiated launch of instance (i-08f5ace73f974eaf8)

Instance summary for i-08f5ace73f974eaf8 (QC-Server) Info

Updated less than a minute ago

Instance ID
i-08f5ace73f974eaf8

IPv6 address
-

Hostname type
IP name: ip-172-16-5-90-ca-central-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
-

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
-

Instance state
 Running

Private IP DNS name (IPv4 only)
ip-172-16-5-90-ca-central-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0c8aa4fc187774f0 (FMCG-VPC)

Subnet ID
subnet-04fb0cb83e8ad007f (Quality Control)

Instance ARN
arn:aws:ec2:ca-central-1:652294806688:instance/i-08f5ace73f974eaf8

Private IPv4 addresses
172.16.5.90

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
 Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role
-

Security groups
sg-0f37423f51e6d6f7 (Private-SG)

Owner ID
652294806688

Launch time
Sat Apr 18 2026 17:23:38 GMT-0600 (Central Standard Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0bf5c50763ce8191e	All	All	172.16.0.0/16	Private-SG	-
-	sgr-0bd95a93ab18a0ca0	22	TCP	sg-042a31c1485b50e6a	Private-SG	-

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sgr-00ef088a5066c07e3	All	All	0.0.0.0/0	Private-SG	-

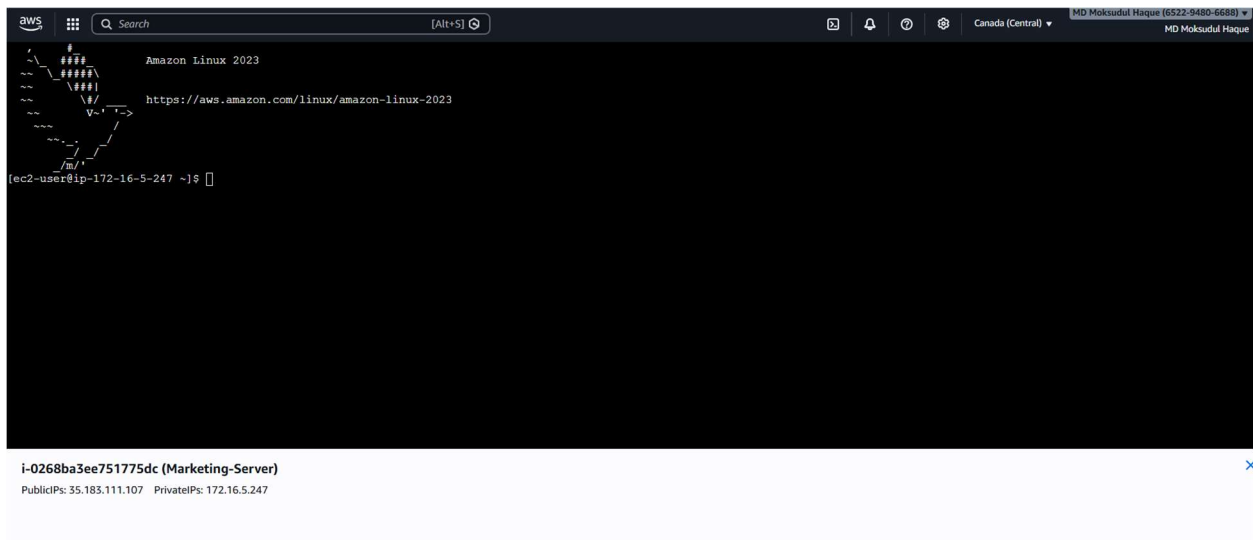
10.0 All the Instances

Instances (9) Info

Last updated less than a minute ago  [Connect](#) [Instance state](#)

Saved filter sets		Find Instance by attribute or tag (case-sensitive)						
Choose filter set								
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	
☐	HR-Server	i-07a56719e351470a2	 Running	t3.micro	 3/3 checks passed	View alarms +	ca-central-1a	
☐	Legal-Server	i-06cbeec50c967cb0d	 Running	t3.micro	 Initializing	View alarms +	ca-central-1a	
☐	Marketing-Ser...	i-0268ba3ee751775dc	 Running	t3.micro	 3/3 checks passed	View alarms +	ca-central-1a	
☐	Sales-Server	i-0a54368e666a0886d	 Running	t3.micro	 3/3 checks passed	View alarms +	ca-central-1a	
☐	QC-Server	i-08f5ace73f974eaf8	 Running	t3.micro	 Initializing	View alarms +	ca-central-1a	
☐	Strategy-Server	i-04a9ce42c5e819178	 Running	t3.micro	 Initializing	View alarms +	ca-central-1a	
☐	Finance-Server	i-04258a87840f53cf8	 Running	t3.micro	 3/3 checks passed	View alarms +	ca-central-1a	
☐	CS-Server	i-024d99cbaaf9f5943	 Running	t3.micro	 3/3 checks passed	View alarms +	ca-central-1a	
☐	Technical-Server	i-088cab98a65d6a78c	 Running	t3.micro	 3/3 checks passed	View alarms +	ca-central-1a	

10.1 Connect to Public EC2 (Marketing-Server)

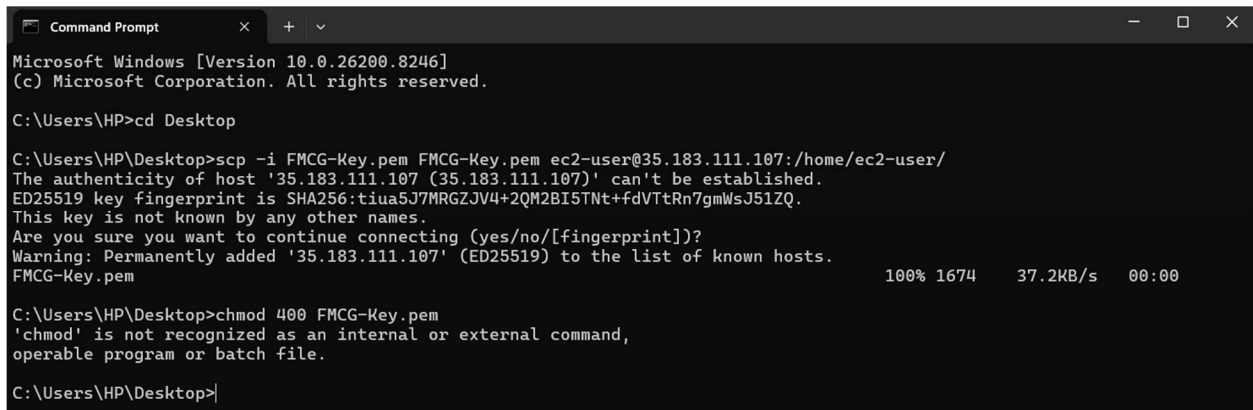


10.2 Connect to Private EC2

Run SCP command in **Run > CMD**

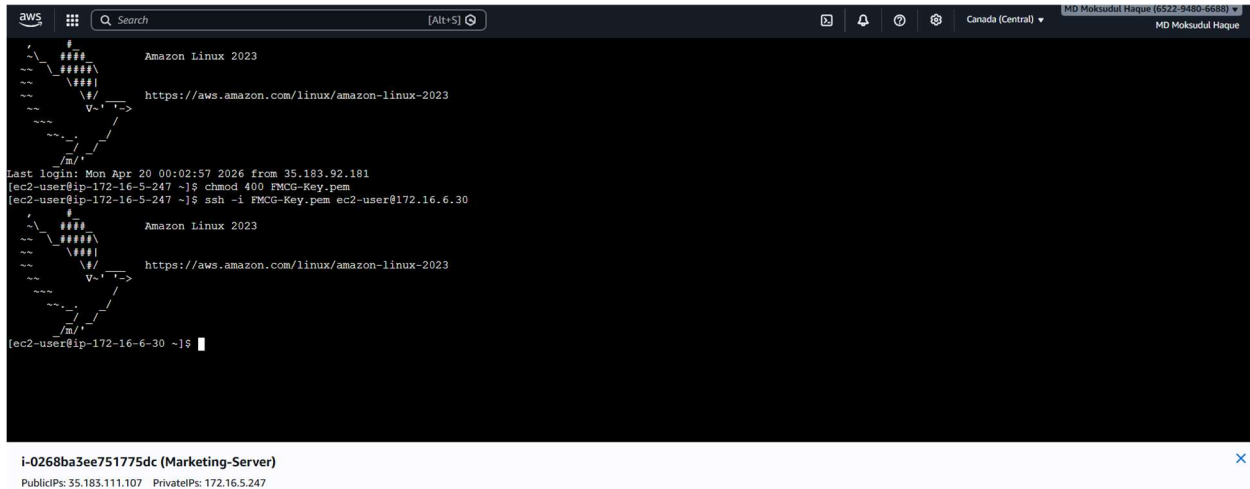
FMCG-Key.pem, located in **Desktop**.

Upload and Run **FMCG-Key.pem** from **Desktop**



10.3 From AWS terminal, Private subnet internet access

Public to Private internet access is working, and Bastion host is working perfectly.



```
aws
Search [Alt+S]
Canada (Central) MD Moksudul Haque (6522-9480-6688) MD Moksudul Haque

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Mon Apr 20 00:02:57 2026 from 35.183.92.181
[ec2-user@ip-172-16-5-247 ~]$ chmod 400 FMCG-Key.pem
[ec2-user@ip-172-16-5-247 ~]$ ssh -i FMCG-Key.pem ec2-user@172.16.6.30

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

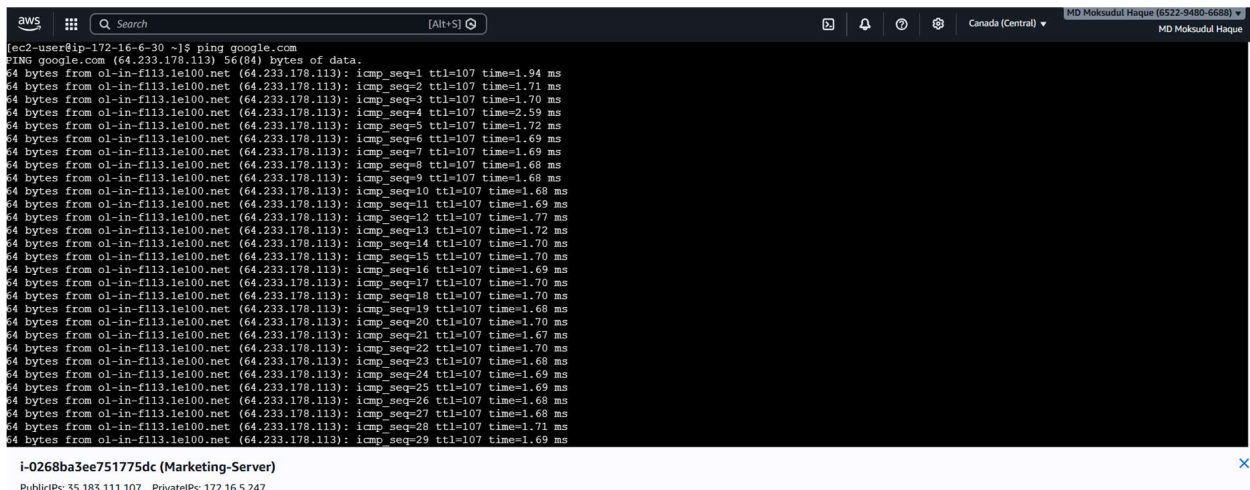
[ec2-user@ip-172-16-6-30 ~]$
```

i-0268ba3ee751775dc (Marketing-Server)
PublicIPs: 35.183.111.107 PrivateIPs: 172.16.5.247

Public to Private connections are working perfectly.

10.4 Test Internet (NAT Gateway Proof)

ping google.com



```
aws
Search [Alt+S]
Canada (Central) MD Moksudul Haque (6522-9480-6688) MD Moksudul Haque

[ec2-user@ip-172-16-6-30 ~]$ ping google.com
PING google.com (64.233.178.113) 56(84) bytes of data.
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=1 ttl=107 time=1.94 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=2 ttl=107 time=1.71 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=3 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=4 ttl=107 time=2.59 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=5 ttl=107 time=1.72 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=6 ttl=107 time=1.69 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=7 ttl=107 time=1.69 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=8 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=9 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=10 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=11 ttl=107 time=1.69 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=12 ttl=107 time=1.77 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=13 ttl=107 time=1.72 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=14 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=15 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=16 ttl=107 time=1.69 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=17 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=18 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=19 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=20 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=21 ttl=107 time=1.67 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=22 ttl=107 time=1.70 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=23 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=24 ttl=107 time=1.69 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=25 ttl=107 time=1.69 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=26 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=27 ttl=107 time=1.68 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=28 ttl=107 time=1.71 ms
64 bytes from ol-in-f113.1e100.net (64.233.178.113): icmp_seq=29 ttl=107 time=1.69 ms
```

i-0268ba3ee751775dc (Marketing-Server)
PublicIPs: 35.183.111.107 PrivateIPs: 172.16.5.247

11. Bastion Host Explanation

11.0 Bastion Host Architecture in AWS

In this architecture, the **Marketing EC2 instance** is configured as a **Bastion Host (Jump Server)** to securely access private EC2 instances located in private subnets.

A Bastion Host is a specially configured EC2 instance that acts as a secure entry point for managing instances in private networks. It is placed in a **public subnet** and has a public IP address, allowing controlled SSH access from the internet.

How it works in our project:

1. The **Marketing-Server EC2 (Public Subnet)** has:
 - Public IP enabled
 - Security Group allowing SSH access only from the user's IP
2. All other EC2 instances (HR, Finance, Sales, etc.) are placed in **private subnets**:
 - No public IP assigned
 - Cannot be accessed directly from the internet
3. Access flow:
 - Laptop (SSH)
 - Marketing EC2 (Bastion Host)
 - Private EC2 Instances (via private IP)

Security Benefits:

- Private EC2 instances are completely hidden from the internet
- Only one controlled entry point exists (Bastion Host)
- Reduces attack surface significantly
- Administrators must authenticate through the public EC2 first

Summary:

This setup ensures a **secure administrative access model**, where only the Bastion Host is exposed publicly, while all backend systems remain protected inside private subnets.

12.0 AWS Cost Calculation (10-Year Estimate Report)

Cost Estimation Model for AWS Architecture

This section compares the long-term operational cost of the AWS-based infrastructure with traditional on-premises solutions.

Assumptions:

- 1 VPC
- 1 NAT Gateway
- 1 Public EC2 (Bastion Host)
- 8–9 Private EC2 instances (t2.micro / t3.micro)
- Continuous running (24/7)

Monthly Cost Breakdown

EC2 Instances

- 1 Public EC2 (Bastion Host): ~\$8–10/month
- 8 Private EC2 instances: ~\$60–70/month total

EC2 Total: ~\$70–80/month

NAT Gateway

- NAT Gateway hourly charge + data processing:
Approximately **\$40–50/month**

Data Transfer & Miscellaneous

- Internal traffic + logs + small outbound usage:
Approximately **\$10/month**

Total Estimated Monthly Cost

Component	Estimated Cost
EC2 Instances	\$75
NAT Gateway	\$45
Data Transfer	\$10
Total	~\$130/month

13. 10-Year Cost Projection

$\$130 \times 12 \text{ months} \times 10 \text{ years} = \text{\$15,600 (approx.)}$

One-Time Manpower (Implementation Cost)

- Deployment effort: **40 hours**
- Hourly rate: **\$60/hour**

Manpower Cost:

$40 \text{ Hours} \times \$60 = \text{\$2,400}$

This includes:

- VPC design and setup
- Subnet configuration (public + private)
- Routing (Internet Gateway + NAT Gateway)
- EC2 deployment (all instances)
- Security Groups and Bastion Host setup
- Testing and validation

Final Total AWS Project Cost (Including Manpower)

Cost Type	Amount
10-Year AWS Operational Cost	\$15,600
One-Time Manpower Cost	\$2,400
Total AWS Project Cost	\$18,000

Comparison with Traditional Infrastructure (Cisco / On-Premises)

Solution	Estimated 10-Year Cost
Cisco On-Premises Setup	~\$118,800
AWS Cloud Architecture	~\$18,600

All cost estimations are based on publicly available pricing from Amazon Web Services and validated using the AWS Pricing Calculator

14.0 References (AWS Pricing Sources (With URLs))

- Amazon Web Services. **Amazon EC2 Pricing**. Available at: <https://aws.amazon.com/ec2/pricing/>
- Amazon Web Services. **Amazon VPC Pricing (NAT Gateway)**. Available at: <https://aws.amazon.com/vpc/pricing/>
- Amazon Web Services. **Data Transfer Pricing**. Available at: https://aws.amazon.com/ec2/pricing/on-demand/#Data_Transfer
- Amazon Web Services. **AWS Pricing Calculator**. Available at: <https://calculator.aws/>

Key Cost Advantage:

- No upfront hardware purchase
- No physical maintenance cost
- Pay-as-you-go pricing model
- Resources can be scaled up/down anytime
- Reduced implementation effort compared to traditional systems

15.0 Conclusion:

AWS provides a **cost-efficient and scalable alternative** to traditional infrastructure. In this design, the organization achieves:

- ~85% cost reduction over 10 years.
- Higher flexibility and scalability.
- Improved security via VPC segmentation.
- Reduced operational and maintenance overhead.

Part 03: Comparison & Recommendation

1.0 Comparison and Decision Factors

To determine which network solution is more suitable for the FMCG organization, both the on-premises Cisco network and the AWS VPC cloud-based network were carefully compared. The comparison was based on key factors relevant to the company's size, future growth, and overall operational needs. These factors include cost, scalability, maintenance, security, flexibility, deployment speed, and reliability.

Category	Cisco On-Premises (Pros)	Cisco On-Premises (Cons)	AWS VPC (Pros)	AWS VPC (Cons)
Cost	One-time hardware investment	Very high upfront cost (~\$118,800)	Low upfront cost, pay-as-you-go (~\$18,000 over 10 years)	Ongoing operational costs
Scalability	Can scale by adding hardware	Scaling is slow and expensive	Highly scalable, resources can be added instantly	Costs increase with usage
Maintenance	Full control over hardware	Requires IT staff and ongoing maintenance	AWS manages infrastructure	Less direct hardware control
Deployment Speed	Stable once installed	Time-consuming setup and configuration	Rapid deployment of resources	Requires cloud knowledge
Security	Full physical control	Manual configuration required	Built-in security features (private subnets, security groups, NAT)	Depends on proper configuration
Reliability	Reliable with proper maintenance	Hardware failure risks	High availability and redundancy	Dependent on internet connectivity
Flexibility	Fixed infrastructure	Difficult to modify	Highly flexible and dynamic	Can become complex at scale
Accessibility	Works without internet internally	Limited remote accessibility setup	Accessible from anywhere via internet	Requires internet access

The decision factors above were selected because they directly impact the organization's operational efficiency and long-term sustainability:

- Cost – evaluates total investment over time
- Scalability – ability to support future growth
- Maintenance – effort required to manage the system
- Security – protection of sensitive data
- Flexibility – ability to adapt to changes
- Reliability – system stability and uptime

2.0 Analysis:

The Cisco on-premises solution is a traditional approach where all networking equipment, such as switches and routers, is physically installed and managed by the organization. This gives full control over the infrastructure, which can be useful in certain situations. However, it comes with several disadvantages. The initial setup cost is very high, and the organization must also spend time and resources maintaining the hardware. In addition, if the company grows in the future, new equipment must be purchased and installed, which increases both cost and deployment time.

On the other hand, the AWS VPC solution provides a modern cloud-based approach. Instead of relying on physical devices, the network is created virtually using AWS services. One of the biggest advantages of AWS is scalability. The company can easily increase or decrease resources depending on demand, which is especially important for departments like Customer Service that have many users and expected growth.

Another major advantage is reduced maintenance. Since AWS manages the underlying infrastructure, the organization does not need to worry about hardware failures, repairs, or upgrades. This allows the company to focus more on its core business activities rather than managing IT infrastructure.

In terms of security, both solutions are capable of providing strong protection. Cisco relies on physical control and manual configurations, while AWS uses built-in features such as private subnets, security groups, and controlled access through NAT and bastion hosts. These features help protect sensitive data while still allowing necessary communication within the network.

Overall, AWS provides a more efficient, flexible, and cost-effective solution for this organization compared to the traditional Cisco approach.

3.0 Final Recommendation:

Based on the comparison and analysis, AWS VPC is recommended as the best network solution for the FMCG organization.

The most significant factor in this decision is cost. The Cisco solution has an estimated total cost of approximately \$118,800 over 10 years, while the AWS solution is around \$18,000. This is a major difference and makes AWS a much more affordable option for the company.

In addition to cost savings, AWS offers better scalability. The organization can easily handle growth without needing to redesign the network or purchase new hardware. This is very important for a growing company, especially with departments like Customer Service that already require many resources.

AWS also simplifies maintenance. Since the infrastructure is managed by AWS, the organization does not need to dedicate as many resources to maintaining and troubleshooting hardware. This reduces workload and operational complexity.

From a security perspective, AWS still provides strong protection. By using private subnets, controlled access, and secure connection methods, sensitive systems can remain protected while still allowing necessary communication.

4.0 Hybrid Consideration

A hybrid solution can also be considered as an alternative approach. In this case, critical departments such as Finance, HR, and Legal could remain on a Cisco on-premises network, while other departments and scalable services are hosted on AWS.

This approach can provide additional control over sensitive data and systems. However, it also introduces more complexity, as the organization would need to manage both on-premises and cloud environments. It may also increase overall cost due to maintaining two different systems and ensuring proper integration between them.

Because of these challenges, while a hybrid approach is a valid option, it may not be the most efficient solution for this organization.

5.0 Final Statement

In conclusion, AWS VPC is the most suitable solution for the FMCG organization. It offers significant cost savings, better scalability, improved flexibility, and reduced maintenance compared to the Cisco on-premises network. Although Cisco provides strong control and reliability, AWS is a more practical and future-ready solution that better meets the needs of the organization.
